

Assignment: Electronic Bandstructure of Bilayer Graphene

April 6, 2026

1 Tight-Binding Model of AB-Stacked Bilayer Graphene

In lecture, we derived the low-energy bandstructure of monolayer graphene using a nearest-neighbor tight-binding model. In monolayer graphene, we defined a 2-atom basis (A, B) and considered the intralayer hopping parameter $t \approx 3.03$ eV (The sign of the hopping term can be set arbitrarily through multiplying a phase factor). This resulted in a 2×2 Hamiltonian $H(\vec{k})$ that yielded the linear dispersion, $E(\vec{q}) = \pm \hbar v_F |\vec{q}|$, where $\vec{q} = \vec{k} - \vec{K}$.

For this problem, we will extend this methodology to derive the bandstructure of AB-stacked (Bernal) bilayer graphene.

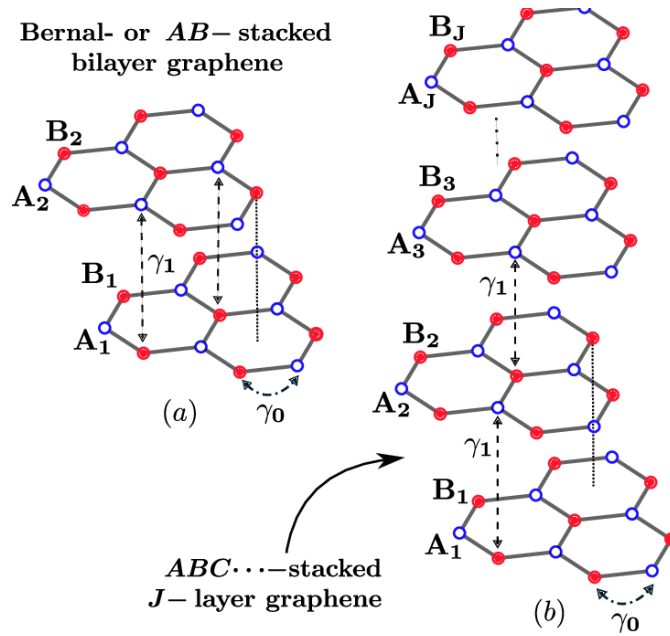


Figure 1: Schematic illustration of the stacking configuration in (a) Bernal-stacked bilayer graphene and (b) ABC-stacked J-layer graphene. Adapted from Dora and Balram, Phys. Rev. B., 2023.

To do this, you need to understand the structural differences and how they modify the Hamiltonian:

- **Expanded Unit Cell:** Your unit cell now contains four atoms including both bottom and top layer. We will define our basis as (A_1, B_1, A_2, B_2) , where subscript 1 denotes the bottom layer and 2 denotes the top layer.
- **Stacking Geometry:** In AB stacking, the top layer is shifted such that the A_2 atoms lie directly above the B_1 atoms. Refer to Figure 1 above.

- **Interlayer Coupling:** We will keep the intralayer hopping $t \approx 3.03$ eV identical to the monolayer case, which we will denote as γ_0 as shown in Figure 1. However, we must introduce a new nearest-neighbor interlayer hopping parameter, $\gamma_1 \approx 0.39$ eV, which couples the vertically aligned A_2 and B_1 orbitals. For this model, ignore all other higher-order couplings and only consider γ_0 and γ_1 .

With these modifications in mind, complete the following steps to derive the low-energy dispersion of bilayer graphene. Refer to the first chapter ('Graphene: basic properties') of the 2D materials book uploaded on Courseworks.

1. **The 4×4 Hamiltonian:** Using the basis (A_1, B_1, A_2, B_2) , construct the 4×4 Hamiltonian matrix $H_{bi}(\vec{k})$. Your matrix should effectively contain two monolayer Hamiltonians on the block diagonal. The off-diagonal blocks will contain the γ_1 coupling. Pay close attention to the order of your basis elements to ensure γ_1 connects B_1 and A_2 and not A_1 - B_2 .

Hint: The structure factor $S(\vec{k})$ will be 1 for the interlayer term, since there is no xy offset between A_2 and B_1 .

2. **Eigenvalues and General Dispersion:** Solve $\det(H_{bi}(\vec{k}) - EI) = 0$ to find the four energy eigenvalues $E(\vec{k})$ in terms of γ_0 , γ_1 , and the structural factor $S(\vec{k})$.

3. **Low-Energy Dispersion:** Focus on the behavior near the Dirac point $\vec{K}$. Substitute the $E(\vec{k}) = \gamma_0 f(\vec{K} + \vec{q}) \approx \hbar v_F |\vec{q}|$ into your energy equation for the two lowest-energy bands (the conduction and valence bands closest to the Fermi level). Plot the energy at $-0.05 \text{ \AA}^{-1} \leq k \leq 0.05 \text{ \AA}^{-1}$.

Hint: Apply the Taylor expansion $\sqrt{1+x} \approx 1 + x/2$. Show that the resulting dispersion consists of two parabolic bands that touch at zero energy. You will be able to find the effective mass m^ from the parabolic dispersion relation $E = \pm \frac{\hbar^2 q^2}{2m^*}$.*

4. **Introducing Gap:** Unlike monolayer graphene, bilayer graphene's bandstructure can be significantly altered by an external perpendicular electric field of voltage V . This field creates an interlayer potential difference of V . The on-site energies of layer 1 are now $-V/2$ and layer 2 are $V/2$. Modify your Hamiltonian from part (1) to include this asymmetry. Solve for the two lowest energy eigenvalues near the K point. Drop all terms higher than 4th order of $\vec{k}$ or $\vec{q}$. Plot the energy at $-0.05 \text{ \AA}^{-1} \leq k \leq 0.05 \text{ \AA}^{-1}$, for $V = 50, 100 \text{ meV}$. Notice the difference between the plots in question 3 and 4.

Hint 1: Use the same approximation ($E(\vec{k}) \approx \hbar v_F |\vec{q}|$) as in question 3.

Hint 2: You would have to apply second order Taylor expansion twice by factoring out $\gamma_1^2/2$ and $V/2$. The expansion formula is $\sqrt{1+x} \approx 1 + x/2 - x^2/8$.

Hint 3: Assume $\hbar v_F |\vec{q}| \ll V \ll \gamma_1$. Thus, drop higher order terms such as V^3/γ_1^6 or V/γ_1^4 that will appear through the derivation.

Example code for solving 2 to 4 (Python):

```

import sympy as sp

r0, S, r1, E, V = sp.symbols('r0 S r1 E V')

H = sp.Matrix([
    [0, r0*S, 0, 0],
    [r0*S, 0, r1, 0],
    [0, r1, 0, r0*S],
    [0, 0, r0*S, 0]
])

I = sp.eye(4)

char_eq = (H - E*I).det()
print(f"Characteristic Equation: {char_eq} = 0")

solutions = sp.solve(char_eq, E)
print(f"Solutions for E: {solutions}\n")

```